AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 2 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- Buffer: $[H^+] = K_a \times [HA] \div [A^-]$ — use moles, the volume cancels
- Acid added to a buffer: $A^- + H^+ \rightarrow HA$, so mol HA goes up and mol A^- goes down by the same amount
- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$
- Pure water is neutral at every temperature because $[H^+] = [OH^-]$; its pH is only 7 at 25 °C

Section A: Short-answer questions

01 Which statement about pH is correct?

[1 mark]

A The pH of a weak base is independent of temperature.
B At temperatures above 298 K, the pH of pure water is less than 7.
C The pH of 2.0 mol dm^{-3} nitric acid is approximately 0.30
D The pH of 0.10 mol dm^{-3} sulfuric acid is greater than that of 0.10 mol dm^{-3} hydrochloric acid.

02 Explain why the pH of a buffer solution does not change when it is diluted with water.

[1 mark]

03 A calibrated pH meter is used to monitor the pH during a titration of hydrochloric acid with sodium hydroxide. Explain why the volume of sodium hydroxide solution added between each pH measurement is made smaller as the end point of the titration is approached.

[1 mark]

04 Which change causes the pH of 10 cm^3 of 1.0 mol dm^{-3} NaOH to be halved at 298 K? ($K_w = 1.0 \times 10^{-14}$ at 298 K)

[1 mark]

A adding 10 cm^3 of water
B adding 10 dm^3 of water
C adding 5 cm^3 of 1.0 mol dm^{-3} HCl
D adding 10 cm^3 of 1.0 mol dm^{-3} HCl

05 State the meaning of the term *weak acid*.

[1 mark]

Section B: Longer questions

06 A student prepared a buffer solution by adding 0.0136 mol of a salt KX to 100 cm^3 of a 0.500 mol dm^{-3} solution of a weak acid HX and mixing thoroughly.

06.1 The student then added 3.00×10^{-4} mol of potassium hydroxide to the buffer solution.

Calculate the pH of the buffer solution after adding the potassium hydroxide. For HX at 25 °C, $K_a = 1.41 \times 10^{-5}$ mol dm^{-3}. Give your answer to two decimal places.

[6 marks]

06.2 A buffer solution has a constant pH even when diluted. Use a mathematical expression to explain this.

[1 mark]

07 The ionic product of water, $K_w = 2.93 \times 10^{-15}$ mol^2 dm^{-6} at 10 °C

07.1 Calculate the pH of pure water at 10 °C. Give your answer to two decimal places.

[2 marks]

07.2 Suggest why this pure water at 10 °C is not alkaline.

[1 mark]

07.3 Calculate the pH of a 0.0131 mol dm^{-3} solution of calcium hydroxide at 10 °C. Give your answer to two decimal places.

[3 marks]

07.4 The 0.0131 mol dm^{-3} calcium hydroxide solution at 10 °C was a saturated solution. A student added 0.0131 mol of magnesium hydroxide to 1.00 dm^3 of water at 10 °C and stirred the mixture until no more solid dissolved. Predict whether the pH of the magnesium hydroxide solution formed is larger than, smaller than or the same as the pH of the calcium hydroxide solution at 10 °C. Explain your answer.

[2 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7405/3 Nov 2020 Q13*	1
02	The ratio [HA] ÷ [A^-] stays the same, so [H^+] = K_a × [HA] ÷ [A^-] is unchanged	*Source: Practice question (written for this set)*	1
03	To avoid missing the end point / (very little pH change at the start but) large change in pH near the end point	*Source: 7405/1 Nov 2021 Q06.6*	1
04	**D**	*Source: 7405/3 Jun 2023 Q17*	1
05	(Acid) partially / slightly ionises or dissociates (in water to form H^+ ions)	*Source: 7405/1 Nov 2020 Q04.1* Allow does not fully ionise / dissociate	1
06.1	M1 $K_a = [H^+][X^-] \div [HX]$ or $[H^+] = K_a \times [HX] \div [X^-]$ M2 n(HX) = 0.0500 mol M3 n(HX) after adding KOH = $0.0500 - 3 \times 10^{-4} = 0.0497$ mol M4 n(KX) after adding KOH = $0.0136 + 3 \times 10^{-4} = 0.0139$ mol M5 $[H^+] = 1.41 \times 10^{-5} \times 0.0497 \div 0.0139 = 5.04 \times 10^{-5}$ M6 pH = 4.30	*Source: 7405/1 Jun 2019 Q09.4–09.5* If no attempt at M3 and M4, max 2 marks Answer to 2 dp	6
06.2	[H^+] = K_a × [HX] ÷ [X^-] **and** the ratio remains (almost) constant on dilution		1
07.1	M1 $[H^+] = \sqrt{K_w} = \sqrt{(2.93 \times 10^{-15})}$ $(= 5.41 \times 10^{-8})$ M2 pH = 7.27	*Source: 7405/1 Jun 2017 Q03.2–03.5* Must be 2 dp. 7.27 scores 2 marks	2
07.2	[H^+] = [OH^-]	Allow description in words, e.g. equal amounts of H^+ and OH^-	1
07.3	M1 $[OH^-] = 0.0131 \times 2 = 0.0262$ M2 $[H^+] = K_w \div [OH^-] = 2.93 \times 10^{-15} \div 0.0262$ $(= 1.118 \times 10^{-13})$ M3 pH = 12.95	12.95 scores 3 marks 12.42 scores 2 ($K_w = 1 \times 10^{-14}$ used) 12.65 scores 1 (not × 2)	3
07.4	M1 smaller / lower pH / less alkaline M2 magnesium hydroxide is less soluble / solubility of hydroxides increases down Group 2	If not smaller, CE = 0/2 Do not allow $Mg(OH)_2$ is insoluble	2